
Beyond Parallel Corpora

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data and machine learning

Supervised and Unsupervised



- We framed machine translation as a supervised machine learning task
 - training examples with labels
 - here: input sentences with translation
 - structured prediction: output has to be constructed in several steps

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 - we will also look at using just sentences output language

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 - some labeled training data
 - some unlabeled training data (usually more)

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- Semi-supervised learning
 - some labeled training data
 - some unlabeled training data (usually more)
- Self-training
 - make predictions on unlabeled training data
 - use predicted labeled as supervised translation data

Transfer Learning



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- Multi-Task training
 - train on a related task first
 - e.g., part-of-speech tagging
- Share some or all of the components

using monolingual data

Using Monolingual Data



- Language model
 - trained on large amounts of target language data
 - better fluency of output
- Key to success of statistical machine translation
- Neural machine translation
 - integrate neural language model into model
 - create artificial data with backtranslation

Adding a Language Model



- Train a separate language model
- Add as conditioning context to the decoder

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- Add as conditioning context to the decoder
- Recall state progression in the decoder
 - decoder state s_i
 - embedding of previous output word Ey_{i-1}
 - input context c_i

$$s_i = f(s_{i-1}, Ey_{i-1}, c_i)$$

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- Pre-train language model
- Leave its parameters fixed during translation model training

Refinements



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- Learn a scaling factor (gate) $\text{gate}_i^{\text{LM}} = f(s_i^{\text{LM}})$

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- Learn a scaling factor (gate) $\text{gate}_i^{\text{LM}} = f(s_i^{\text{LM}})$
- Use it to scale values of language model state

$$\bar{s}_i^{\text{LM}} = \text{gate}_i^{\text{LM}} \times s_i^{\text{LM}}$$

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$$\bar{s}_i^{\text{LM}} = \text{gate}_i^{\text{LM}} \times s_i^{\text{LM}}$$

- Use this scaled language model state for decoder state

$$s_i = f(s_{i-1}, Ey_{i-1}, c_i, \bar{s}_i^{\text{LM}})$$

backtranslation

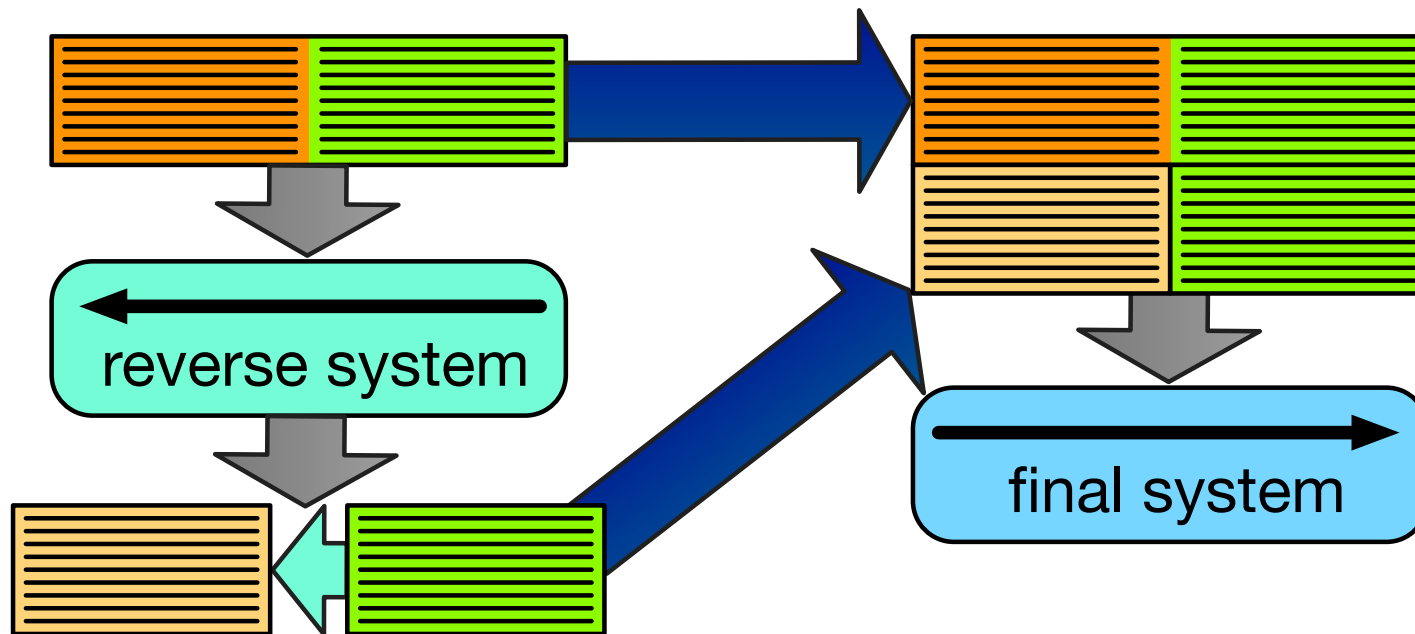
Back Translation



- Monolingual data is parallel data that misses its other half

Back Translation

- Monolingual data is parallel data that misses its other half
- Let's synthesize that half



- Steps
 1. train a system in reverse language translation
 2. use this system to translate target side monolingual data
→ synthetic parallel corpus
 3. combine generated synthetic parallel data with real parallel data to build the final system
- Roughly equal amounts of synthetic and real data
- Useful method for domain adaptation

Iterative Back Translation



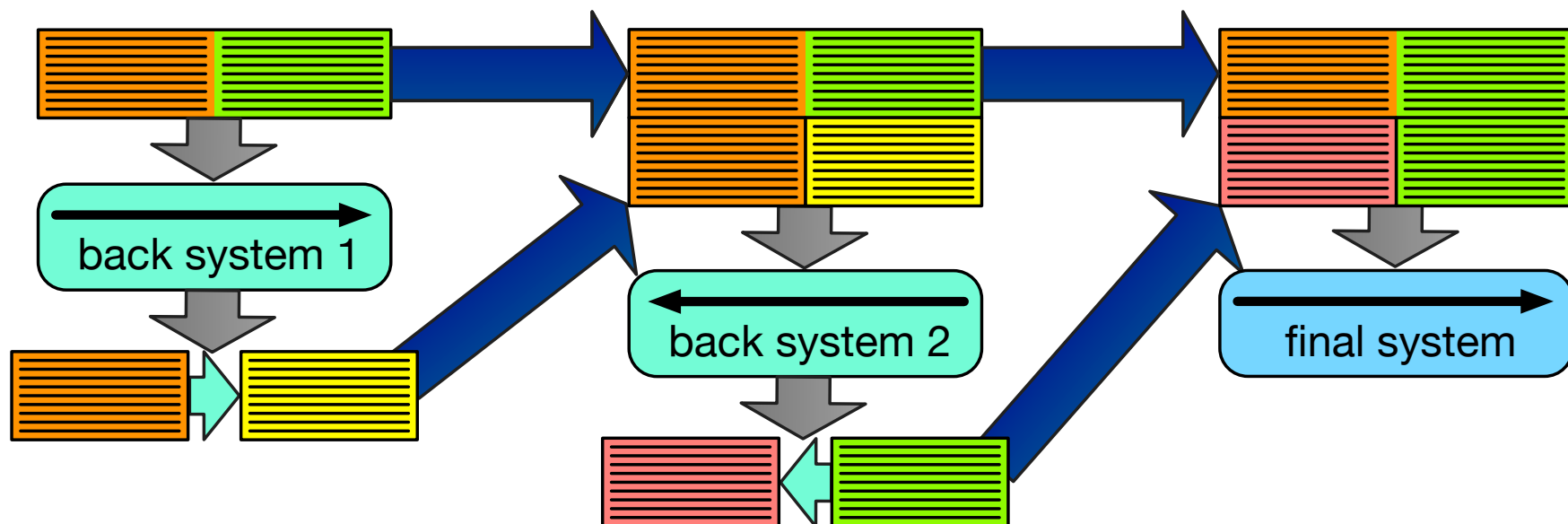
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Iterative Back Translation

- Quality of backtranslation system matters
- Build a better backtranslation system ... with backtranslation

Iterative Back Translation

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Iterative Back Translation

- Example: Better system for backtranslation matters

German–English	Back	Final
no back-translation	-	29.6
*10k iterations	10.6	29.6 (+0.0)
*100k iterations	21.0	31.1 (+1.5)
convergence	23.7	32.5 (+2.9)
re-back-translation	27.9	33.6 (+4.0)

* = limited training of back-translation system

- Copy Target
 - if no good neural machine translation system to start with
 - just copy target language text to the source
- Forward Translation
 - synthesize training data in same direction as training
 - self-training (inferior but sometimes successful)

Round Trip Training

- We could iterate through steps of
 - train system
 - create synthetic corpus

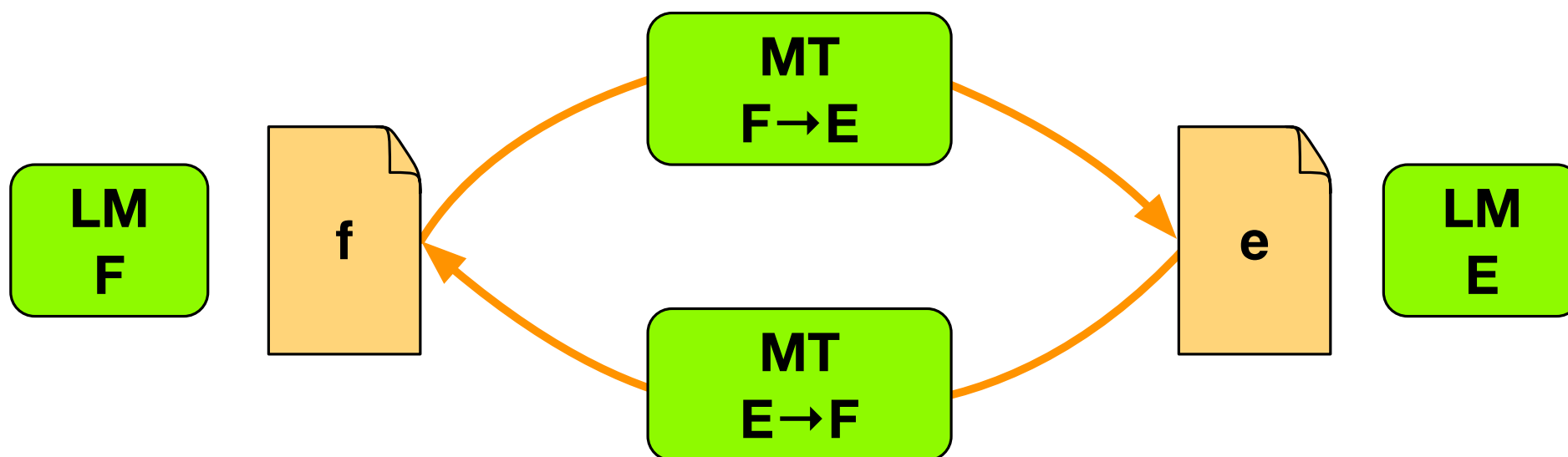
Round Trip Training

- We could iterate through steps of
 - train system
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- Dual learning: train models in both directions together
 - translation models $F \rightarrow E$ and $E \rightarrow F$
 - take sentence **f**
 - translate into sentence **e'**
 - translate that back into sentence **f'**
 - training objective: **f** should match **f'**

Round Trip Training

- We could iterate through steps of
 - train system
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- Dual learning: train models in both directions together
 - translation models $F \rightarrow E$ and $E \rightarrow F$
 - take sentence $\mathbf{f}$
 - translate into sentence $\mathbf{e}'$
 - translate that back into sentence $\mathbf{f}'$
 - training objective: $\mathbf{f}$ should match $\mathbf{f}'$
- Setup could be fooled by just copying ($\mathbf{e}' = \mathbf{f}$)
 - $\Rightarrow$ score $\mathbf{e}'$ with a language for language E
 - add language model score as cost to training objective

Round Trip Training



monolingual pretraining

Low Resource Language Pairs

- Problem: not enough parallel to even train a proper encoder or decoder
- Idea: use monolingual data
 - ... in source language → initialize encoder
 - ... in target language → initialize decoder
- How do we present monolingual data in training?

Masked Training

- Replace some input word sequences with <pad> (30% of words)
- Train model MASKED → TEXT on both source and target text

Why did the chicken cross the road?

↑

Why did <pad> chicken <pad> the road?

Reordering Sentences

- Reorder sentences (each training example has 3 sentences)

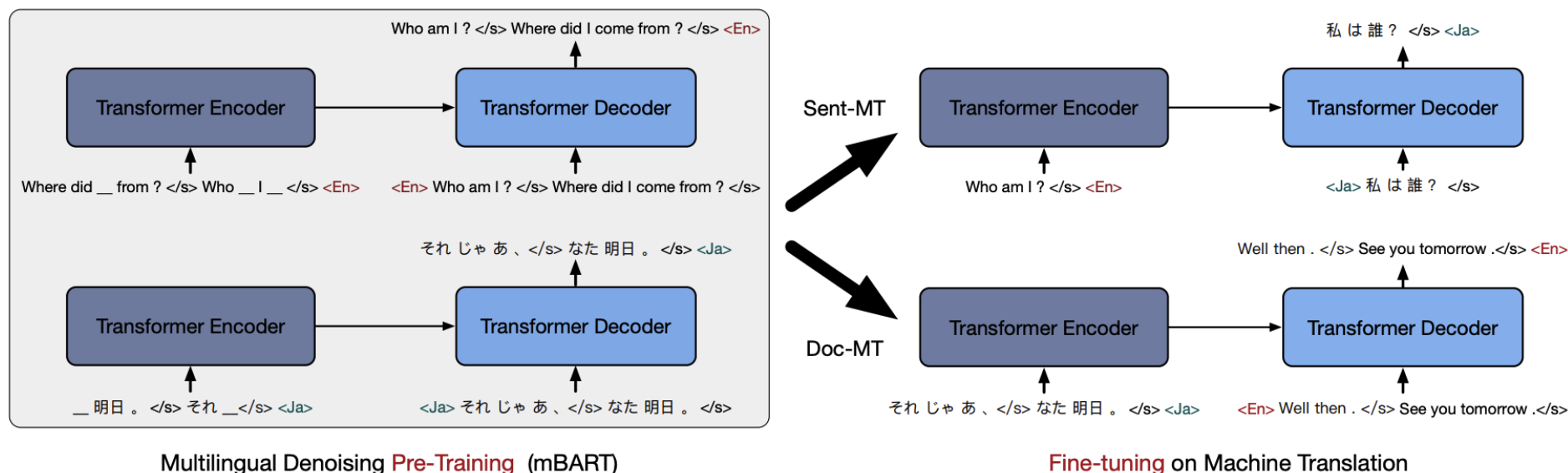
*Why did **the** chicken **cross** the road?*
*The chicken wanted to get **to the other side**.*
*There are some delicious **sunflower** seeds.*

⇓

*The chicken wanted to get **<pad>** other **<pad>**.*
***<pad>** are some delicious **<pad>** seeds.*
*Why did **<pad>** chicken **<pad>** the road?*

Example: mBART

“Multilingual Denoising Pre-training for Neural Machine Translation” (Liu et al., 2020)

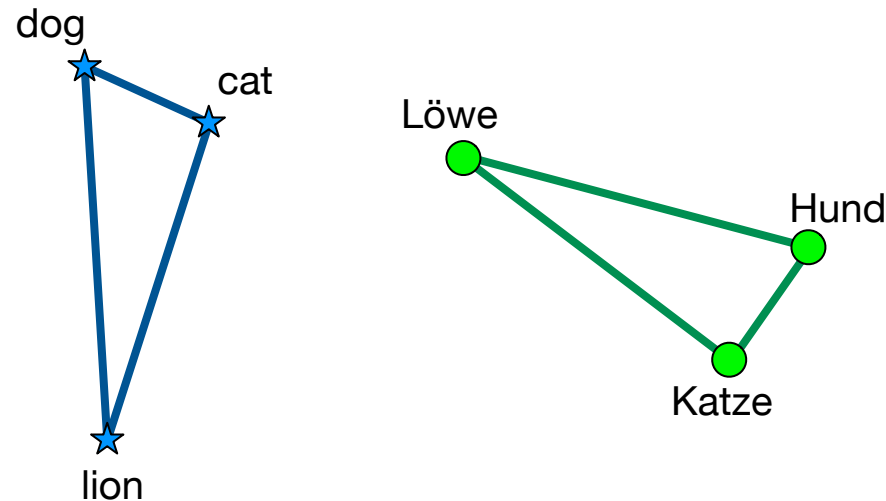


- 25 languages: from 55 billion words English to 56 million words Burmese
 - Followed by training on parallel data
- ⇒ Helps with low-resource languages
(but not with >20 million sentence pair parallel data)



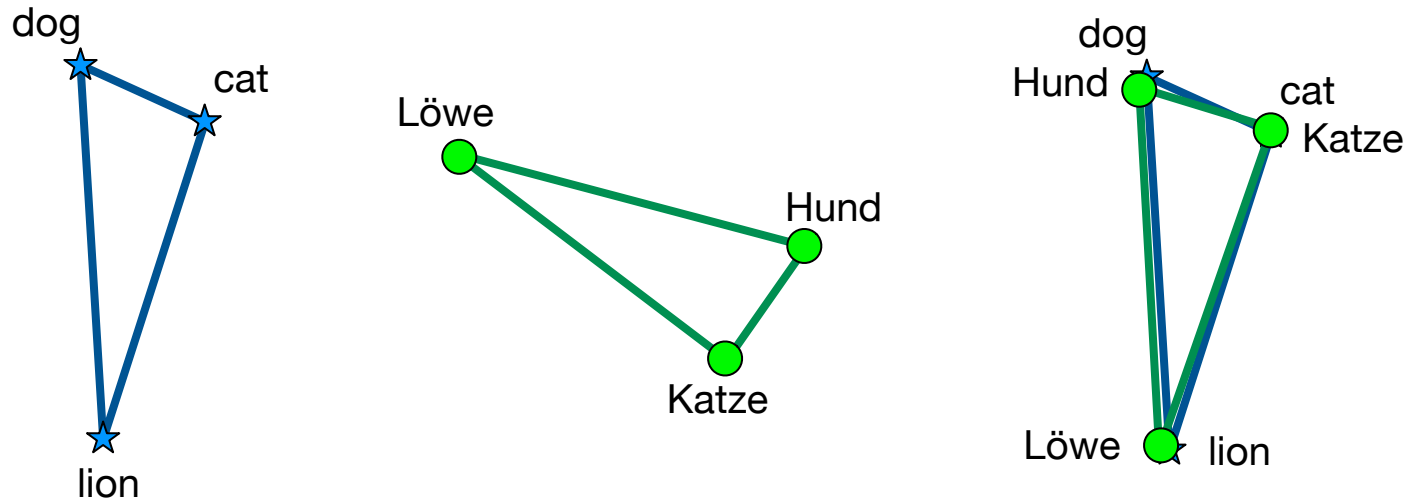
unsupervised machine translation

Monolingual Embedding Spaces



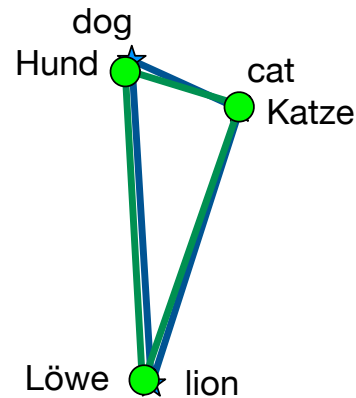
- Embedding spaces for different languages have similar shape
- Intuition: relationship between *dog*, *cat*, and *lion*, independent of language
- How can we rotate the triangle to match up?

Matching Embedding Spaces



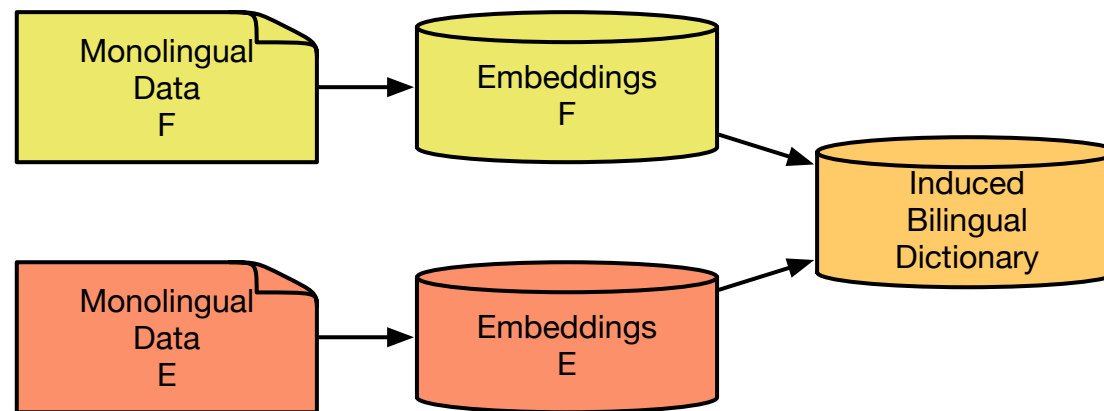
- Seed lexicon of identically spelled words, numbers, names
- Adversarial training: discriminator predicts language [Conneau et al., 2018]
- Match matrices with word similarity scores: Vecmap [Artetxe et al., 2018]

Bilingual Lexicon Induction



- Given shared embedding state

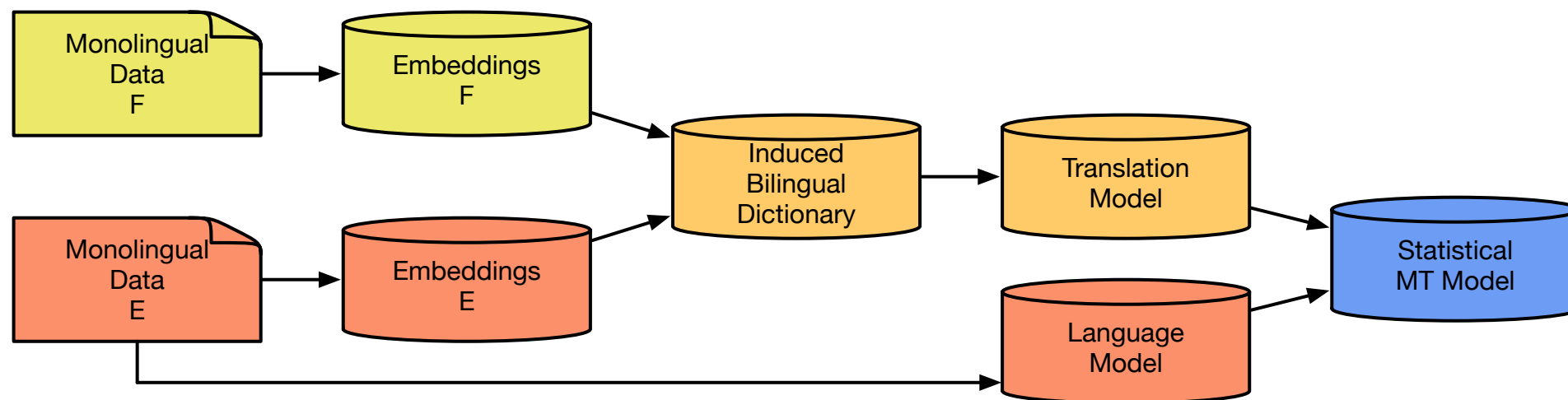
⇒ matching points in space = word translations



Inferred Translation Model

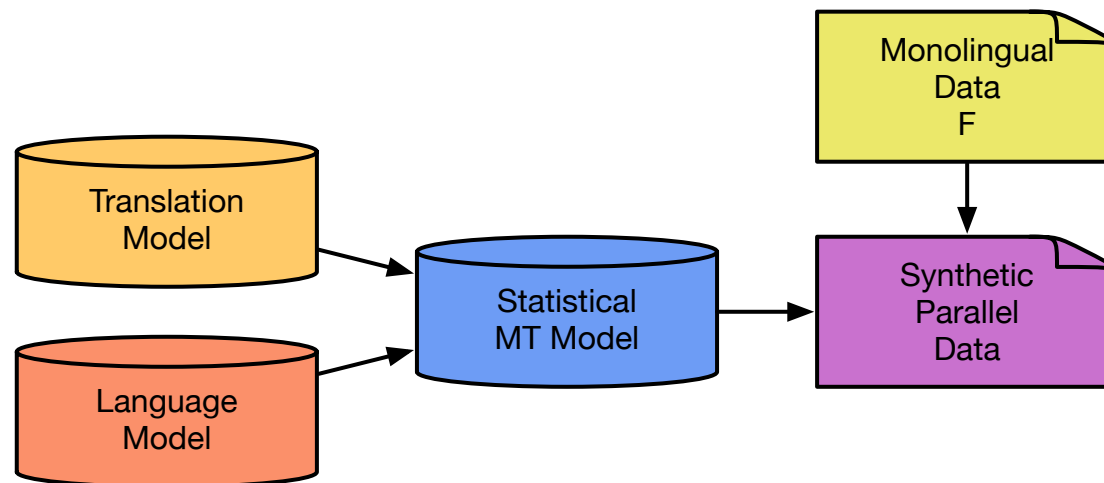
- Translation model
 - induced word translations
 - statistical phrase translation table (probability $\simeq$ similarity)
- Language model
 - target side monolingual data
 - estimate statistical n-gram language model

⇒ Statistical phrase-based machine translation system

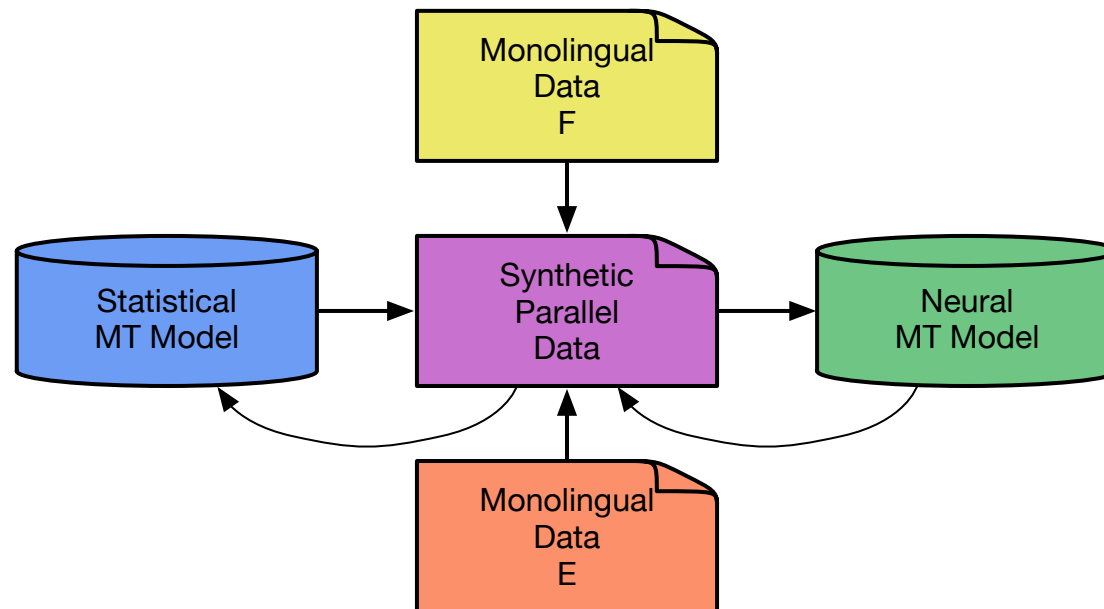


Synthetic Training Data

- Create synthetic parallel corpus
 - monolingual text in source language
 - translate with inferred system: translations in target language



- Iterate
 - Predict data: generate translation for monolingual corpus
 - Predict model: estimate model from synthetic data
 - iterate this process, alternate between language directions
- Increasingly use neural machine translation model to synthesize data



multiple language pairs

Multiple Language Pairs

- There are more than two languages in the world
- We may want to build systems for many language pairs
- Typical: train separate models for each
- Joint training

Multiple Input Languages

- Example
 - German–English
 - French–English
- Concatenate training data
- Joint model benefits from exposure to more English data
- Shown beneficial in low resource conditions
- Do input languages have to be related? (maybe not)

Multiple Output Languages

- Example
 - French–English
 - French–Spanish
- Concatenate training data
- Given a French input sentence, how specify output language?

Multiple Output Languages

- Example
 - French–English
 - French–Spanish
- Concatenate training data
- Given a French input sentence, how specify output language?
- Indicate output language with special tag

[ENGLISH] *N'y a-t-il pas ici deux poids, deux mesures?*

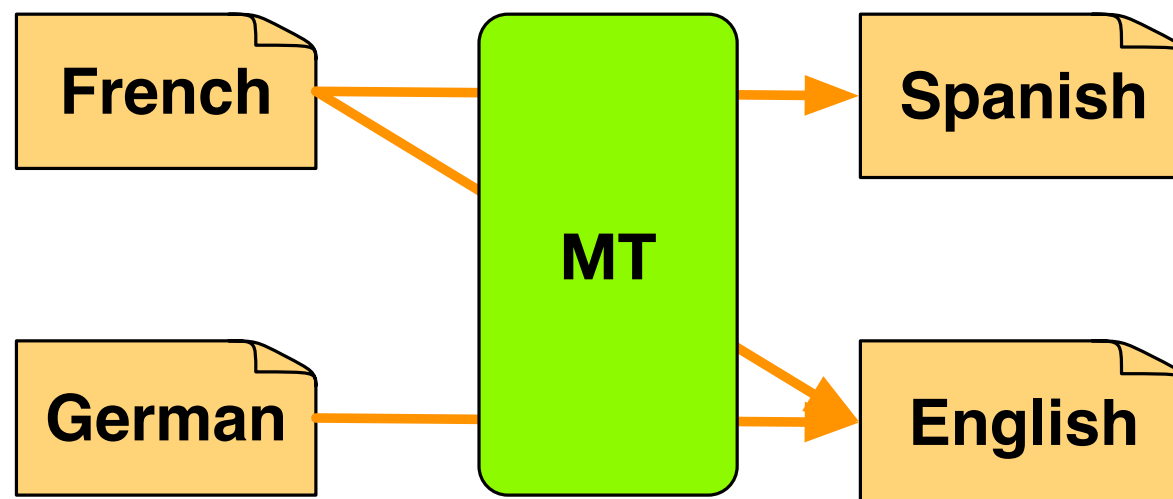
⇒ *Is this not a case of double standards?*

[SPANISH] *N'y a-t-il pas ici deux poids, deux mesures?*

⇒ *¿No puede verse con toda claridad que estamos utilizando un doble rasero?*

Zero Shot Translation

- Example
 - German–English
 - French–English
 - French–Spanish
- We want to translate
 - German–Spanish



- Train on
 - German–English
 - French–English
 - French–Spanish
- Specify translation

[SPANISH] *Messen wir hier nicht mit zweierlei Maß?*

⇒ *¿No puede verse con toda claridad que estamos utilizando un doble rasero?*

Algorithms

Google's AI just created its own universal 'language'

The technology used in Google Translate can identify hidden material between languages to create what's known as interlingua

By **MATT BURGESS**

23 Nov 2016

Table 5: Portuguese→Spanish BLEU scores using various models.

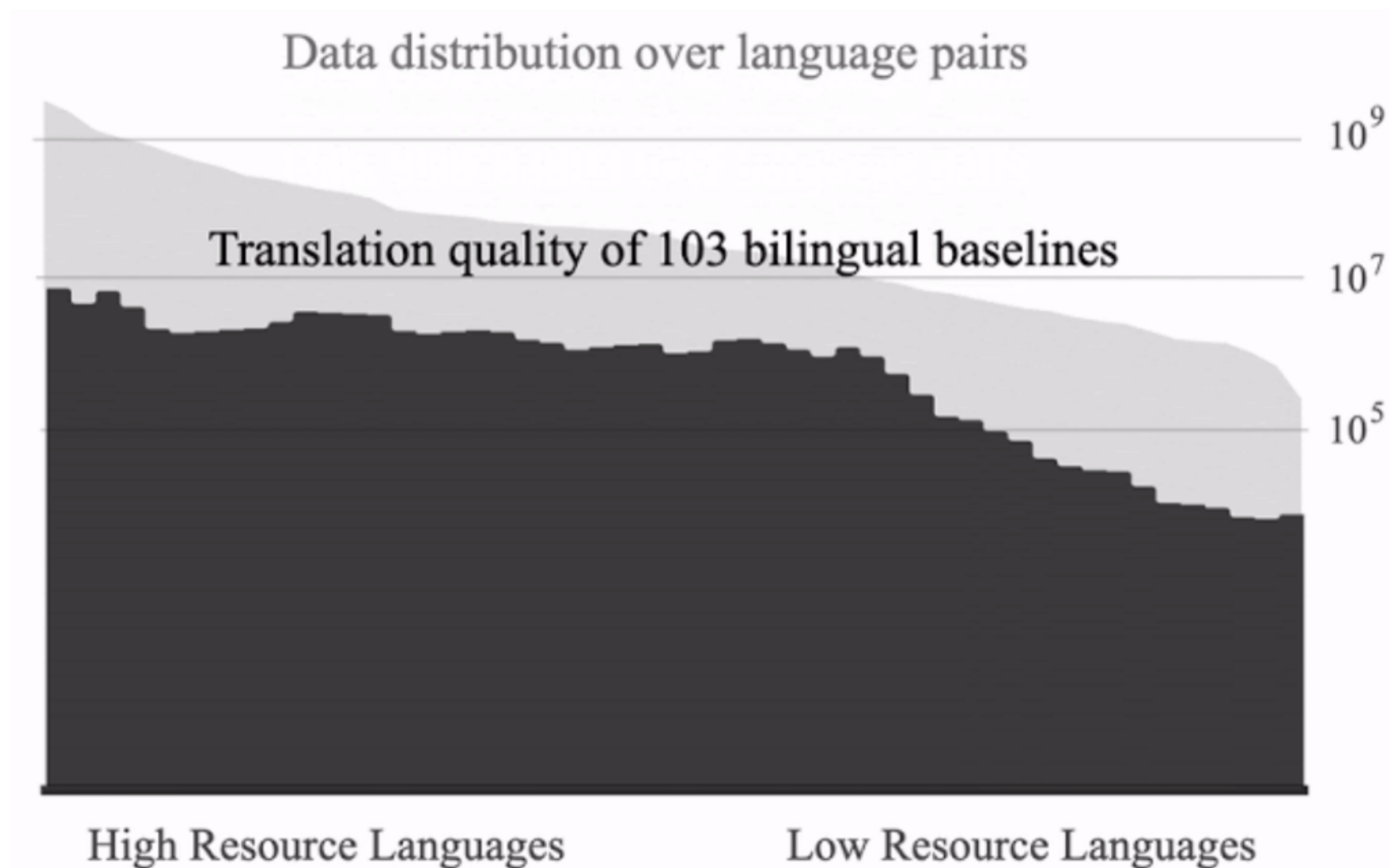
	Model	Zero-shot	BLEU
(a)	PBMT bridged	no	28.99
(b)	NMT bridged	no	30.91
(c)	NMT Pt→Es	no	31.50
(d)	Model 1 (Pt→En, En→Es)	yes	21.62
(e)	Model 2 (En↔{Es, Pt})	yes	24.75
(f)	Model 2 + incremental training	no	31.77

- Bridged: pivot translation Portuguese → English → Spanish
- Model 1 and 2: Zero shot training
- Model 2 + incremental training: use of some training data in language pair

Massively Multilingual Training

- Scaling up multilingual machine translation for more languages
 - many-to-English
 - English-to-many
 - many-to-many
- Mainly motivated by improving low-resource language pairs
- Move towards larger models

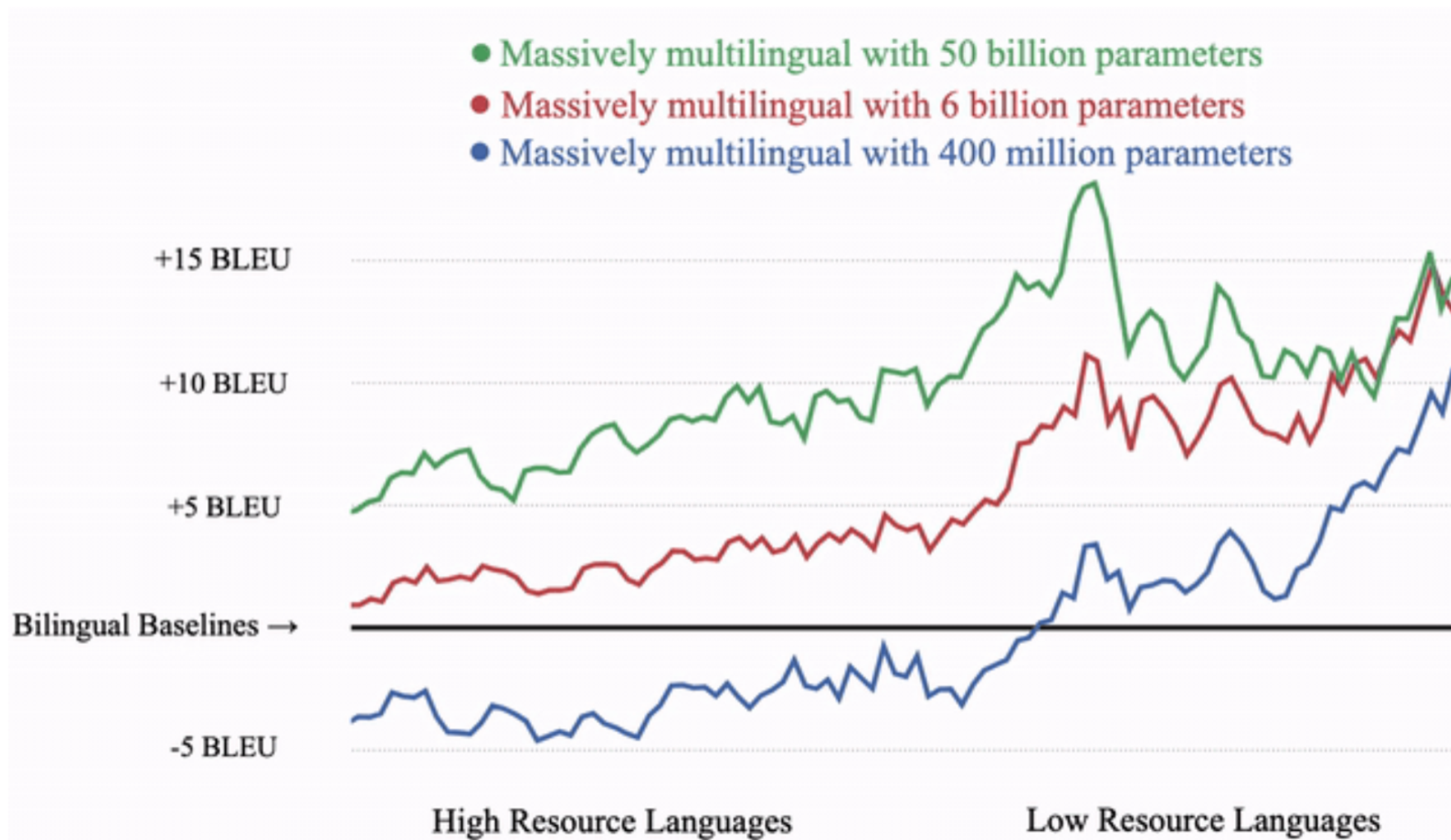
Translation Quality for 103 Languages



(source: Google)

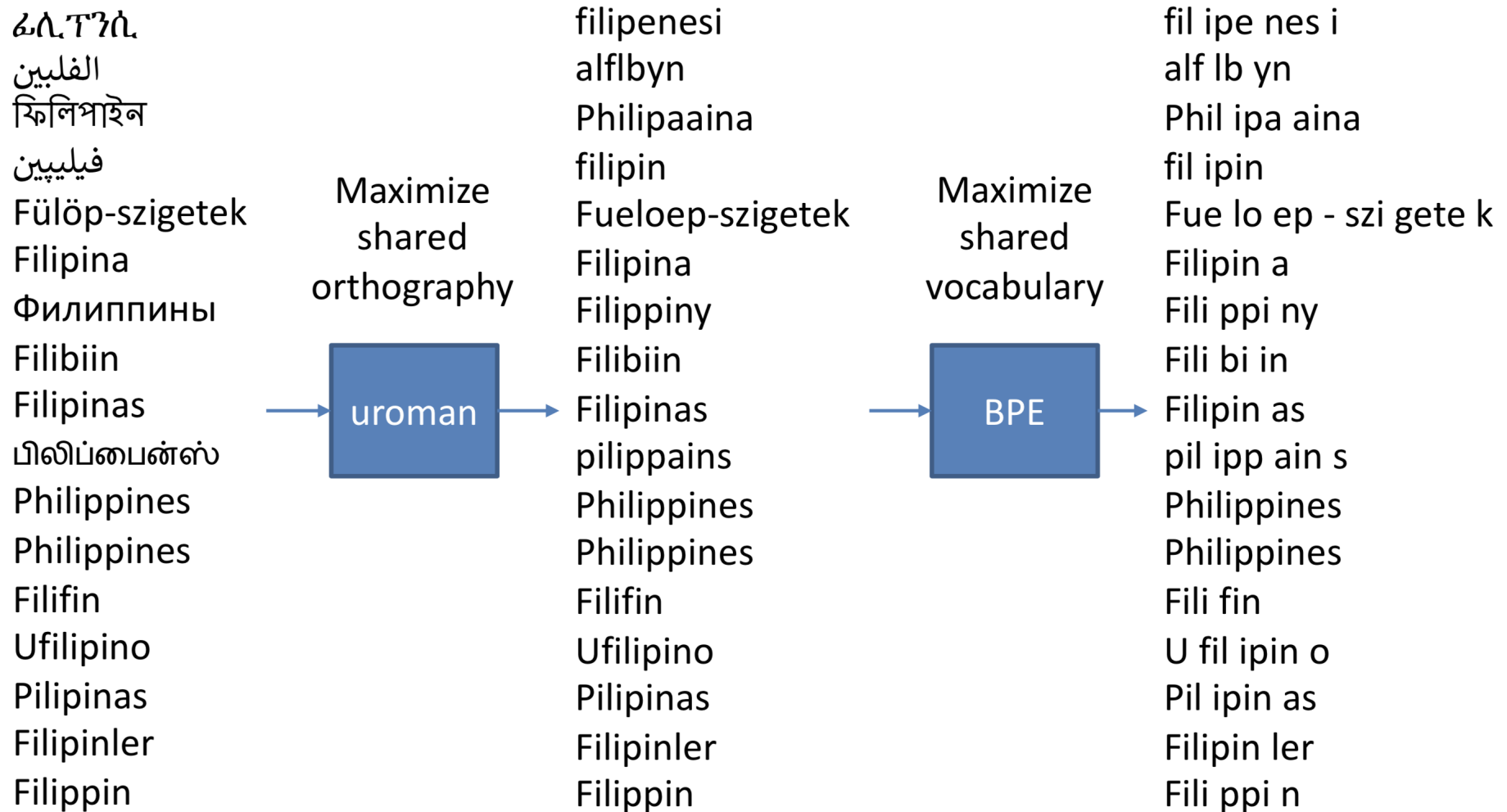
Gains with Multilingual Training

38



(source: Google)

Romanization



(source: USC/ISI)

Facebook

Introducing the First AI Model That Translates 100 Languages Without Relying on English

October 19, 2020

By Angela Fan, Research Assistant

- 7.5 billion sentences for 100 languages (mined from web-crawled data)
- Model with 15 billion parameters
- Improvements especially for low resource languages

Even Bigger: NLLB (2022)

- No Language Left Behind: 200 languages
- Hand-translated test set: Flores-200
- Uses diverse data sources
 - public parallel data
 - translations created by professional translators
 - sentence pairs based on sentence embedding similarity
 - monolingual data for
 - * monolingual pre-training
 - * back-translation
 - * self-training
- Models of different scale (up to 54B parameters), publicly released

Different Amounts of Data per Language

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- High-resource language pairs are undertrained
- Low-resource language pairs are overtrained

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⇒ Oversampling low resource language pairs

Data selection probability p_l for language pair l based on corpus sizes D_k

$$p_l = (D_l / \sum_k D_k)^{1/T}$$



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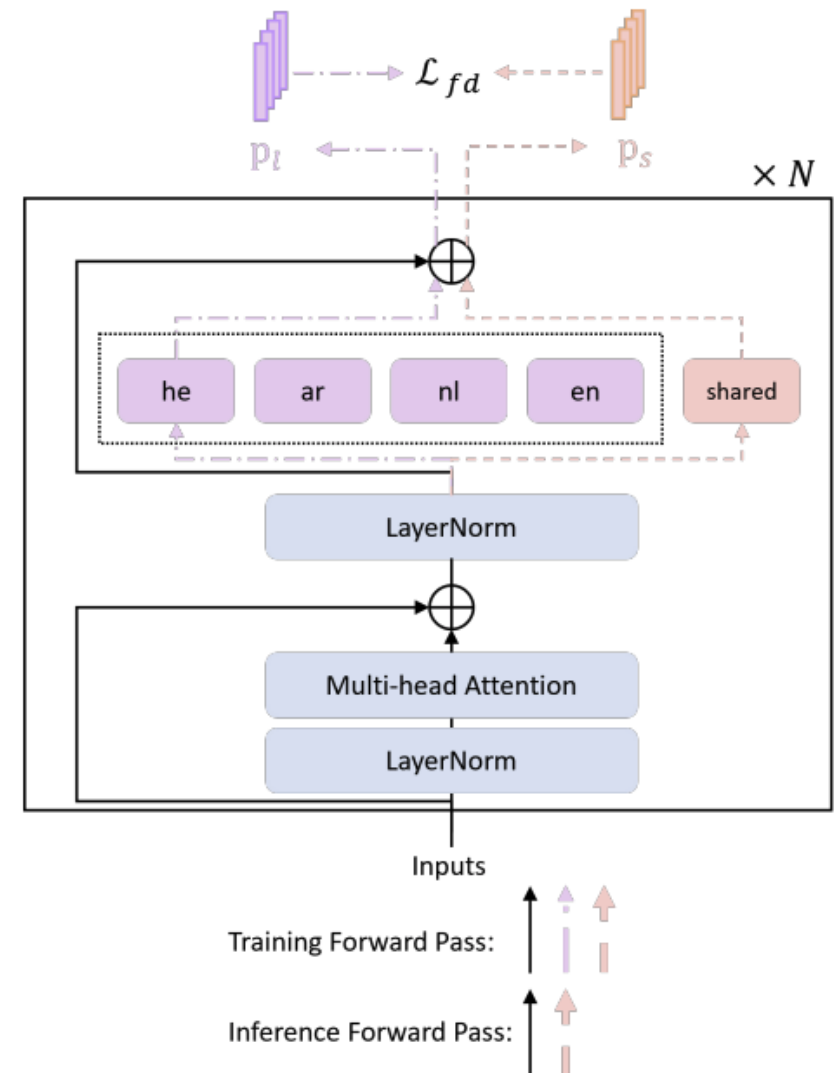
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- Curriculum training: adding low-resource data only in later training stages

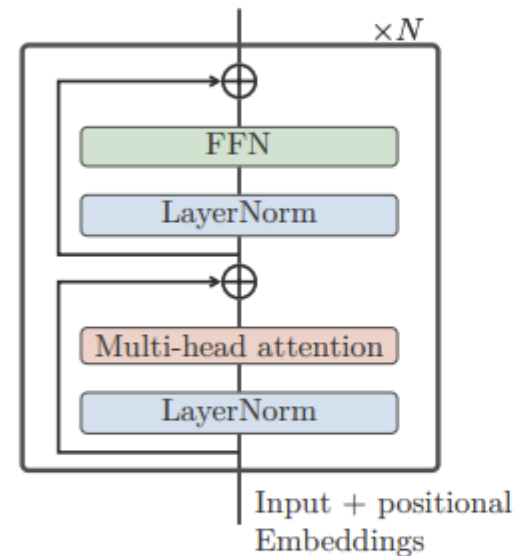
- Many languages in the same representation space
- Beneficial: shared cognates, numbers, names, ...
- Harmful: a lot of accidental overlap in tokens that have different meaning
 - *die* — common German determiner
 - *die* — different meaning in English
- What can be done to avoid harmful interference?

- Various design choices
 - language-specific encoder
 - language-specific decoder
 - language specific adaptor components
- Example:
“Condensing Multilingual Knowledge with Lightweight Language-Specific Modules”
Xu et al. (2023)
 - language specific parameters
 - shared parameters
 - self-distillation method to condense everything into shared parameters

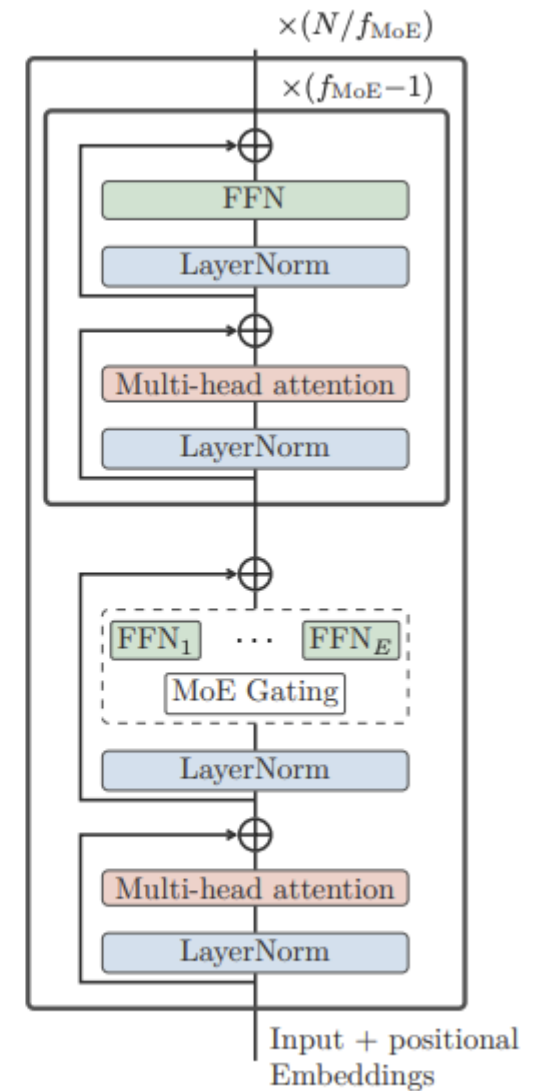


Mixture of Experts

- Conditional compute
- Gating mechanism decides which FF step to utilize
- Allows scaling to many more parameters without increasing computational cost



(a) Dense Transformer



(b) MoE Transformer

document-level translation

The Importance of Document-Level Context ⁴⁷



- Pronouns
 - *I bought a table. It is pretty.*
 - *Ich kaufte einen Tisch. Er/sie/es ist schön.*
- Better disambiguation
 - *I have a lot of numbers. I still need to make the table.*
- Terminological consistency

Why Not Document-Level Translation?



- Entire infrastructure focused on sentence level
 - Training data available as sentence pairs
 - Metrics defined at sentence level
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questions?